

- build 4 sine waves
- each signal is sampled for 10 seconds at 1000 Hz, so each signal has 10,000 samples.
- add random noise to the signals, then create a mixed signal by summing the noisy signals.
- the model input is basically: one hot selector C+ context window from the noisy mixed signal
- the model target is: the aligned clean window from the chosen pure sine wave
- the context window should have 10 samples
- example: choose one of the 4 target sine waves, pick a random location, take 10 noisy mixed samples as input, and the corresponding 10 clean samples from the chosen sine as output.
- we need to implement 3 models: fully connected, RNN, and LSTM
- train them with MSE loss
- the point is to compare the models and explain the results.